Adjoint Shape Optimization of a Conformal Patch Antenna

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Abstract

We design the copper of a probe-fed 2.4 GHz patch antenna on a 1.6 mm FR-4 board bent to a 100 mm radius with geode-fem, a finite-element solver whose discrete shape adjoint reuses one factorization per frequency. We measure cost in EM solves (one frequency, one sparse LU factorization) and count the solves an optimizer spends until the worst $|S_{11}|$ over the band first reaches a target level on the coarse optimization model. With 38 shape parameters the adjoint reaches −9.5 dB in 35 solves; the fastest separable CMA-ES, full CMA-ES or NEWUOA run to reach that level needs 385 (11×, or 8.3× in geode-fem wall time, since an adjoint evaluation costs 1.32 forward ones), and the fastest to reach −9 dB needs 230 against 30 (7.7×, full CMA-ES, one seed). At −9.5 dB the ratio is 2.4×, 2.9× and 11× with 4, 14 and 38 parameters. L-BFGS with a finite-difference gradient follows the adjoint's iterates at 39 times its solves, behind CMA-ES, so the savings come from the adjoint. Re-centred using Palace at third order and a rim-refined mesh, the design covers 2.400–2.4835 GHz at −10 dB on the three most refined models computed, with 1.9 MHz of lower-edge margin on the finest; we claim coverage on those models only. A larger air margin moves the resonance up by 4.3 MHz and further refinement moves it up again, so we estimate that the mesh-limit design misses the lower band edge by 1 to 13 MHz.

1 Introduction

Gradient-based inverse design is routine in nanophotonics, where differentiable solvers such as CEVICHE (Hughes et al., 2019), the Meep adjoint solver (Hammond et al., 2022), SPINS-B (Su et al., 2019) and FDTDX (Mahlau et al., 2026) optimize a permittivity density on a structured grid. Shape derivatives of scattering parameters for finite-element models with ports are older (Georgieva et al., 2002; Nikolova et al., 2004a; Webb and Akel, 2000) and are available in commercial tools (Ansys, Inc., 2025; COMSOL AB, 2023). A gradient is worth its implementation cost only if it buys something that gradient-free search does not. With two nearly independent parameters, finite differences or a scaling law are cheap. With many coupled parameters, finite differences cost one solve per parameter and gradient-free search slows with dimension (Rios and Sahinidis, 2013), which is where an adjoint should pay.

This paper measures that on one antenna. A 2.4 GHz probe-fed patch is bent around a cylinder of radius $R_b = 100$ mm. The board is a uniform 1.6 mm FR-4 shell that is not designed; up to 38 smooth parameters move the copper outline, the corners and the feed. geode-fem, a differentiable finite-element solver with a discrete shape adjoint, optimizes the band objective by L-BFGS. The adjoint is implemented at first element order, so optimization runs on a coarse model (L1, Section 2.1) and the finished design is verified, and re-centred by a fitted frequency shift, on finer models (Sections 5 and 6). On the same problem, objective, start and volume guard we run separable and full-covariance CMA-ES (Hansen and Ostermeier, 2001; Ros and Hansen, 2008) and NEWUOA (Powell, 2006;

Table 1: Names used in the text, with where each is shown.

Name	Meaning	Shown in
Fixture	The whole simulation setup as meshed: board, copper, air box, UPML and port	Fig. 1a
Patch, ground	The copper sheet on top of the board (the only part designed) and the copper on its underside (fixed)	Fig. 1b, c
Board, substrate, slab	The 1.6 mm FR-4 shell between patch and ground, bent to $R_b = 100$ mm; the slab is the substrate under the patch	Fig. 1b
Patch edges $\pm y$, $\pm s$	The four sides of the patch. The $\pm y$ edges end the resonant length and radiate; the $\pm s$ edges run across the bend and do not	Fig. 1b, c, e
Rim	The region within 0.5 mm of the patch edge, where the field is singular and the mesh is refined	Fig. 1d; Table 4
Probe, feed, port	The 1.6 mm-wide connection from ground to patch near the $+y$ edge, modelled as a lumped port; the inset is its distance from that edge	Fig. 1c, e
Air margin, UPML	The air gap between the board and the absorbing layer (UPML, uniaxial perfectly matched layer) that stands in for open space	Fig. 1a
Sagitta	How far the patch edge lies below the patch's crown because of the bend	Fig. 1b
Chart (s, y)	Coordinates on the board: s is arc length across the bend, y runs along the bend axis	Fig. 1e, f
Mesh morphing, volume guard	Moving mesh nodes instead of remeshing; the guard rejects a step that would nearly invert an element	Section 2.4, 2.5
Dip	The frequency of minimum $\lvert S_{11}\rvert$, where the antenna is best matched (the resonance as seen at the port)	Fig. 2
-10 dB band, band edges, margins	The range where $\lvert S_{11}\rvert \le -10$ dB; its ends (the -10 dB crossings); their distances outside the ISM band edges	Fig. 2
Worst-of-band $\lvert S_{11}\rvert$, level	The largest $\lvert S_{11}\rvert$ over the band points (the five objective points while optimizing, Table 2; the nine verification points in Table 6); a level is a target value for it	Fig. 2, 3
EM solve, solves to target	One frequency of the model (one LU factorization); solves spent until a target level is first reached. A *driven solve* is a forward solve as the solver logs it, gradient checks included (the count in Table 5)	Section 1; Fig. 3
L1, P2/L1, refinement levels	The coarse optimization mesh; L1 at element order $p = 2$; L1 with a finer slab, rim or both	Section 2.1; Table 4
Space mapping, δ	Optimizing on L1 for a band shifted down by δ, the fitted coarse-to-fine frequency offset	Section 5
Designs A, B, C	The ends of the design chain; C is the final design	Table 5; Fig. 1f

Zhang et al., 2023) over a range of settings, the same L-BFGS with a finite-difference gradient, and, for two optimizers, the independently developed solver Palace (Palace developers, AWS Center for Quantum Computing, 2023) as the forward model.

The cost measure. We measure an optimizer's cost in *EM solves*. One EM solve is the finite-element model at one frequency: one sparse LU factorization of the 150,561-unknown system of the coarse model and its back-substitutions. The objective samples five frequencies, so one evaluation of it is five solves; on one host it takes 72.1 s, and 95.1 s when the adjoint gradient is computed with it, since the adjoint reuses each factorization for one more back-substitution and then assembles the shape gradient. The quantity we compare is *solves to target*: the number of EM solves an optimizer has spent when its best design so far first reaches a target level of the worst-of-band $\lvert S_{11}\rvert$, the largest reflection over the band (Figure 2 shows how these quantities are read off a response curve). Solves rather than seconds are the measure because they do not depend on the host or on how a run is parallelized; where the adjoint's higher cost per solve matters, we also give the ratio in wall time.

Three results follow.

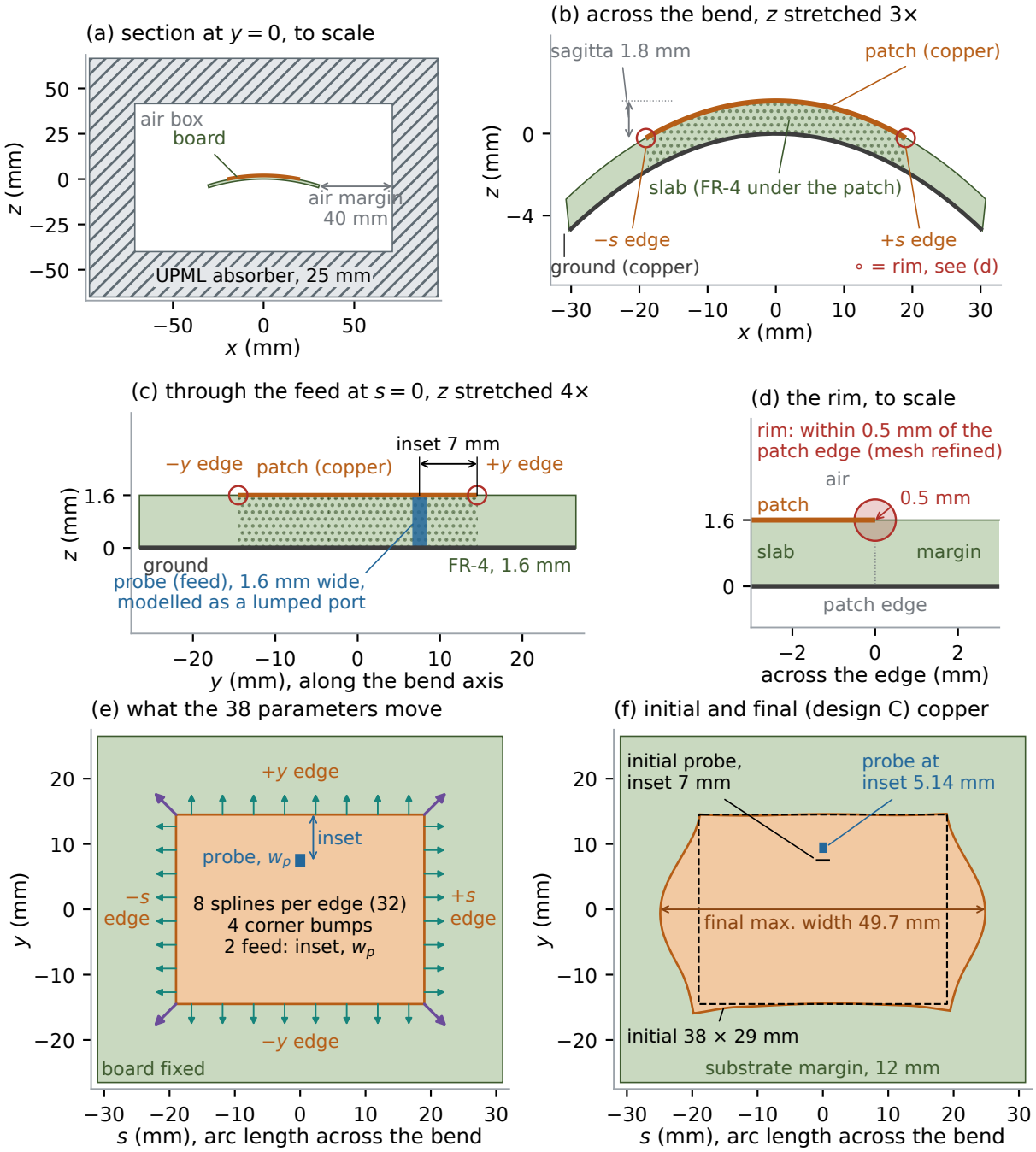


Figure 1: The fixture and its named parts (Table 1). (a) The board inside the air box and the UPML, to scale. (b) Across the bend: the slab is the substrate under the patch (dotted); circles mark the rim. (c) Through the feed: the probe at its initial inset. (d) The rim, the region within 0.5 mm of the patch edge whose mesh Table 4 refines. (e) What the 38 parameters move on the initial patch: each small arrow is one B-spline coefficient (positions schematic), the diagonal arrows the corner bumps, plus the probe's inset and width w_p. (f) The initial patch (dashed) and design C (filled). The board is the same in every design. Drawn from `baseline_2g4_L1.toml` and `recentre/geometry/rbG_L1_copper_38_sm57_w16.json`.

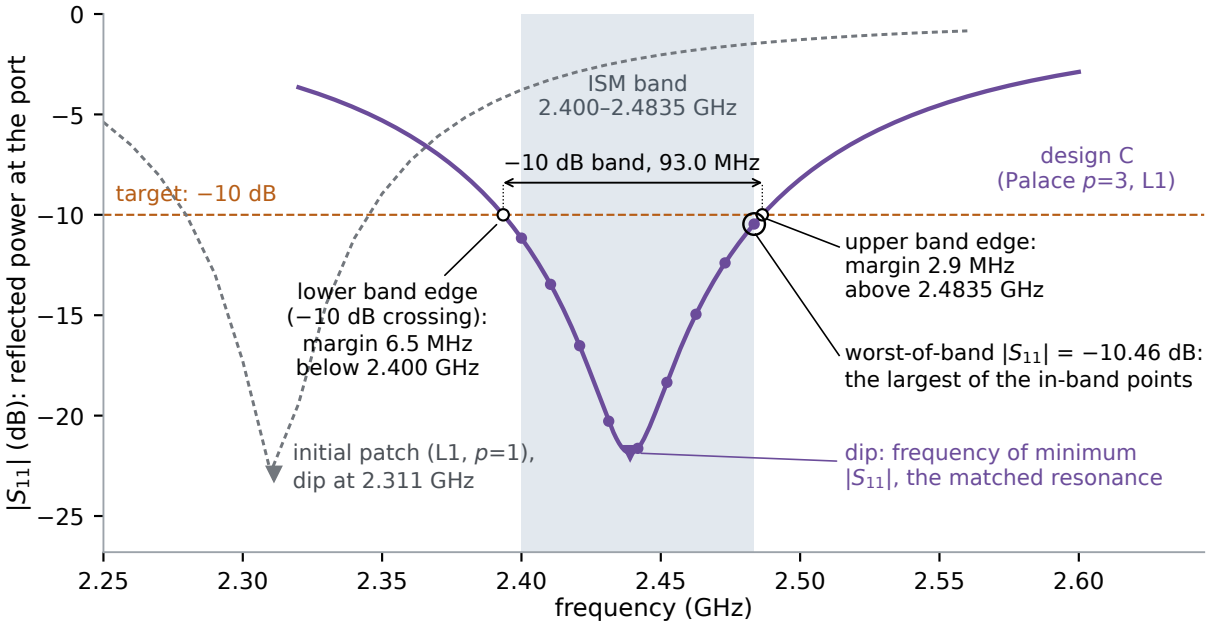


Figure 2: How a return-loss curve is read, and the names the text uses for its features. $|S_{11}|$ is the fraction of incident power reflected at the port, in dB (−10 dB: 10% reflected). Shown are design C (Section 5) on Palace at $p = 3$ on L1 (solid) and the initial patch on L1 at $p = 1$ (dashed). The dip (▼) is the frequency of minimum $|S_{11}|$. The −10 dB band runs between the two −10 dB crossings, the band edges; the margins are how far the band edges lie outside the ISM band (shaded). The worst-of-band $|S_{11}|$ is the largest value over the in-band points (here the nine verification points; the objective uses five), and a level is a target value for it. The initial patch's dip lies 89 MHz below the band, and its worst-of-band value is near −1.5 dB. Data: `baseline_2g4_L1.toml`, `recentre/palace/sm57_p3/port-S.csv`, `recentre/recentre_summary.toml`.

- **The adjoint reaches each target in fewer EM solves, and the ratio is larger with more parameters** (Section 3). To −9.5 dB the fastest gradient-free run needs 2.4×, 2.9× and 11× the adjoint's solves with 4, 14 and 38 parameters, and to −9 dB 1.1×, 3.3× and 7.7×. The adjoint's count to −9.5 dB falls (85, 40, 35) while the best gradient-free count does not follow a trend (205, 115, 385); three sizes of one problem show the direction, not a rate. With 4 parameters three gradient-free runs end lower than the adjoint. A finite-difference gradient on the same L-BFGS path costs 39 times the adjoint's solves, so the savings are the adjoint's, not the gradient's as such; given enough solves, gradient-free search reaches the same level.

- **The forward model's error sits at the patch rim** (Section 4), the thin region along the copper's edge (Figure 1d). Palace on geode-fem's own meshes agrees with geode-fem at first and second order, and at third order with geode-fem's finest rim-refined mesh, and a port-free eigenmode places the matched dip (the frequency of minimum $|S_{11}|$) 16 MHz above the cavity resonance.

- **A design re-centred on the refined model covers the ISM band on the most refined models computed, and is estimated to miss it in the mesh limit** (Section 6). Measured: on the three most refined models its −10 dB band covers 2.400–2.4835 GHz, with 1.9 to 6.6 MHz to spare at 2.400 GHz. Estimated: a larger air margin and further refinement move the band up, and we estimate that the mesh-limit design misses 2.400 GHz by 1 to 13 MHz. We therefore claim ISM coverage for those discrete models, not for the mesh-limit

model or a built antenna. The probe keeps a realizable width (1.6 mm, modelled as a lumped port; an intermediate design had widened it to 4.27 mm, Section 5).

Evidence tiers. We keep three tiers apart. We reserve *bounded* for a proven two-sided bound and use it for no result in this paper; every result here is either *model-dependent*, a value computed for a stated discrete model (mesh, element order, absorber and port model), or *estimated*, an extrapolation to the mesh limit, given with its band and never called a converged value.

Names. The paper refers to parts of the antenna, features of its response and stages of the design by name. Table 1 collects them, each with the figure or table that shows it.

Position of this work. What this paper adds is a measurement on an open-radiator problem (an antenna radiating into free space rather than a closed waveguide) with a moving feed: the solve cost of a finite-element shape adjoint against gradient-free optimizers and a finite-difference gradient at 4, 14 and 38 parameters. We do not claim a first EM shape adjoint (Section 7 places the work), and we make no performance comparison with structured-grid methods.

2 Method

2.1 The fixture

Antenna. The initial antenna is a 38×29 mm PEC patch (width along s, resonant length along y) on 1.6 mm FR-4 with a 12 mm substrate and ground margin, fed by a 1.6 mm probe, modelled as a uniform lumped port, inset 7.0 mm from the $+y$ edge (Figure 1c). The bend axis is parallel to y, so the curvature runs along the non-radiating width; the sagitta at the patch edge is 1.8 mm (Figure 1b). A 40 mm air margin separates the board from the UPML (Figure 1a). The target is $|S_{11}| \leq -10$ dB over the 2.400–2.4835 GHz ISM band.

The coarse mesh. The optimization mesh L1, level 1 of the fixture's uniformly refined mesh family L0 to L3, has 150,561 edges, hence 150,561 unknowns at $p = 1$, a median of 2.0 element layers through the slab and a mean edge length of 0.78 mm at the patch rim. On L1 at $p = 1$ the initial patch's $|S_{11}|$ dip (Figure 2) is at 2.3118 GHz; at $p = 2$ on L1 (P2/L1) it is at 2.3566 GHz, 45 MHz higher. A direct LU solve at P2/L1 (804,460 unknowns) takes 132 GiB with geode-fem's direct solver, so refining further at $p = 2$ is not practical with this solver.

2.2 The open-radiator forward model

The forward problem is the driven frequency-domain curl-curl equation, discretized with Nédélec edge elements of order $p = 1$ or $p = 2$ on an unstructured tetrahedral mesh. Lengths are in millimetres and the solver works with $\omega = k_0 = 2\pi f/c$ in rad/mm. The assembled system is the complex-symmetric pencil

$$A(\omega)\,\mathbf{x} = \Big[\, K(\nu) - \omega^2 M(\varepsilon) + \tfrac{j\omega}{Z_s}\, S_p \,\Big]\,\mathbf{x} = \mathbf{b}, \tag{1}$$

with curl-curl stiffness K, mass matrix M and lumped-port surface term S_p. The radiating boundary is a box UPML of thickness $t = 25$ mm: per-element tensor weights Λ enter K (as $\nu = \Lambda^{-1}$) and M (as $\varepsilon = \varepsilon_r \Lambda$), with the stretch $\varsigma = 1 - j\sigma_0 (d/t)^2/\omega$ per axis at depth d and $\sigma_0 = 1$. The FR-4 has $\varepsilon_r = 4.4$ and $\tan\delta = 0.02$, and the conductors are perfect electric conductors (PEC). The port is a

rectangle of width w_p along y and length $l_p = 1.6\,\mathrm{mm}$ across the substrate, with sheet impedance $Z_s = R_0\, w_p/l_p$ and $R_0 = 50\,\Omega$. The reflection coefficient follows from the port voltage V:

$$S_{11} = \frac{Z - R_0}{Z + R_0}, \qquad Z = \frac{V}{I}, \qquad I = \frac{2V_{\mathrm{inc}} - V}{R_0}. \tag{2}$$

Every evaluation checks passivity, $|S_{11}| \le 1$.

2.3 Shape adjoint with one factorization per frequency

The design moves mesh node coordinates X, so the gradient needs $\partial K/\partial X$, $\partial M/\partial X$, $\partial S_p/\partial X$ and $\partial \mathbf{b}/\partial X$. geode-fem evaluates its closed-form element kernels in first-order dual arithmetic (forward-mode automatic differentiation) inside a tensor stack (Tracel AI, 2024): seeding the tangent of one node coordinate returns the exact partial derivatives of every local matrix entry. The UPML weights are held fixed under node motion. Because $A(\omega)$ is complex symmetric, the adjoint system uses the same operator, and one LU factorization per frequency serves the forward and the adjoint solve. The nodal gradient is $\lambda^{\top}(\partial A/\partial X)\,\mathbf{x}$, with λ the solution of that adjoint system, plus the explicit dependence of the port drive and of the port-voltage functional on the geometry. When a parameter moves the feed, the port faces move and the port is rebuilt on every moved mesh. A unit test of the moving-feed gradient on a curved mesh, with a source that varies across the port so that the explicit term matters, matches central finite differences to below 10^{-8}; without the explicit term it misses by a relative error of about 3.

2.4 Copper design freedom on a fixed curved board

Mesh morphing. The tetrahedral mesh conforms to the copper: the patch outline is made of mesh edges. A design is not remeshed. It moves the mesh nodes instead, which is called mesh morphing: nodes on and under the copper move with it, the air nodes around them follow smoothly, and the absorber and the outer boundary stay fixed. Because the node positions are smooth functions of the design parameters, the assembled system, and with it $|S_{11}|$, is differentiable in them, and the adjoint of Section 2.3 gives that derivative. Morphing has a limit: where a moving edge approaches a fixed feature, the elements between them are squeezed, and a tetrahedron squeezed to zero volume turns inside out (inverts) and invalidates the discretization. The volume guard of Section 2.5 keeps every design short of that.

The morph. The board is not a design variable. Each substrate node has chart coordinates (s, y), with $s = R_b\,\mathrm{atan2}(x, z - z_{\mathrm{axis}})$ the mid-plane arc length and y the bend axis (Figure 1e). A design applies a chart displacement $(\Delta s, \Delta y)$ and moves the node to angle $\phi_0 + \Delta s/R_b$ and height $y_0 + \Delta y$ at its own radius, so the FR-4 stays a uniform 1.6 mm shell. Every run records the largest change of any substrate node's radius ($\le 1.4 \times 10^{-14}\,\mathrm{mm}$). Air nodes follow a harmonic extension of the motion; the UPML and the outer boundary are fixed. The design gradient chains the nodal adjoint gradient through the exact Jacobian of this map. On the refined meshes of Section 6 the air is moved on cylinders instead, since the default morph inverts elements on the combined mesh; on the 0.2 mm-rim mesh the two air motions agree to 0.09 MHz at the lower edge.

The parameters. The 38 parameters (Figure 1e), each per unit coefficient, are 8 clamped cubic B-splines per patch edge ($\pm y$, $\pm s$), each moving its edge outward in the surface by 1 mm, ramped in from the feed band (the strip of the $+y$ edge around the probe) and decaying over the 12 mm substrate margin (32); one bump per corner, moving it 1 mm outward along the bisector with support $1 - \mathrm{smoothstep}(d/6\,\mathrm{mm})$ (4); and the probe's inset from the $+y$ edge and its width w_p, each

1 mm per unit (2). They are smooth functions of the chart, so the same coefficient vector defines the same copper on every mesh of the family, which lets a design be carried between discretizations. The basis cannot open a hole in the patch. For the scaling study two smaller bases are used: 14 parameters (two linear outline splines per edge, the corners and the feed) and 4 (patch width and length, probe inset and probe width); they are separate bases, not subsets of the 38.

2.5 Objective, optimizer and guards

The objective is a soft maximum of $|S_{11}|^2$ over $n_f = 5$ band frequencies, equally spaced from 2.400 to 2.4835 GHz,

$$G(\mathbf{c}) = \frac{1}{\beta} \ln \sum_f \exp\big(\beta\, |S_{11}(f;\mathbf{c})|^2\big), \qquad \beta = 25, \tag{3}$$

over the coefficient vector $\mathbf{c}$. G exceeds the largest $|S_{11}|^2$ by at most $\ln 5/\beta = 0.064$, and every level reported in this paper is the worst of the band points, not G (Figure 2). L-BFGS (memory 8) with Armijo backtracking and a cap of 1 coefficient unit per step minimizes G.

The volume guard and how a descent ends. Each trial step of the line search is first morphed and checked: if it would shrink any tetrahedron's signed volume below 0.25 of its volume in the unmorphed mesh, the step is rejected without a solve and halved. A volume ratio of 0.25 leaves a margin before inversion at 0, so the guard bounds how far the mesh is allowed to be morphed. A descent *ends on the guard* when no trial step both passes the guard and lowers G enough within 25 halvings while some trial failed the guard, or when it reaches a plateau with the most compressed element within 5% of the guard. It *ends on a plateau* when four accepted steps in a row each lower G by less than 5×10^{-4} of its value. Every adjoint descent reported here ends on the guard, except one intermediate run of the space-mapping chain (Section 5), which ended on a plateau. A descent that ends on the guard has stopped where this mesh can be morphed no further along its path, not necessarily at a stationary point of the design problem; the guard's site and what would relieve it are given in Sections 5 and 6.

Before and after each run the pipeline gradient is compared with a central finite difference along the steepest-descent direction and along random directions. The worst relative error over the runs reported here is 1.5×10^{-5} (tolerance 5×10^{-3}), and a per-group check on the two coarsest meshes passes to 1.9×10^{-7}. The adjoint is implemented for $p = 1$ only, so optimization runs at $p = 1$ and finer models enter through verification.

3 The Adjoint Against Gradient-Free Search

Setup. Every optimizer solves one problem: the objective (3) on the unshifted band, the 0.25 volume guard and the L1 fixture at $p = 1$, from $\mathbf{c} = 0$ (a cold start, -1.47 dB). The finite-difference control runs the adjoint's L-BFGS, line search, step cap and guard with a one-sided difference gradient (step 10^{-5}), one band evaluation per parameter. Separable CMA-ES is geode-fem's own implementation of Ros and Hansen (2008), not cross-checked against a reference implementation on a test function; full-covariance CMA-ES is pycma (Hansen et al., 2026) with its defaults. Both use a population of $4 + \lfloor 3 \ln n \rfloor$ samples (14 at $n = 38$), start from an initial step size σ_0, and rank a sample that fails the guard after every feasible sample, by its violation, without a solve. NEWUOA runs in PRIMA with trust radii 0.5 to 10^{-3} and $n + 2$ or $2n + 1$ interpolation points, and a design that fails the guard receives a penalty above any feasible objective, without a solve. For Palace, geode-fem applies the morph and the guard and writes the moved mesh, and Palace solves the five band frequencies at $p = 1$ with GMRES on 16 MPI ranks, a configuration not tuned for speed.

Table 2: EM solves to reach fixed worst-of-band levels (target values of the worst $|S_{11}|$ over the five band points) with 38 parameters on the L1 optimization model ($p = 1$), from $\mathbf{c} = 0$. Slashes separate seeds; the two $\sigma_0 = 0.5$ separable CMA-ES rows use the same three seeds in the same order and coincide to about evaluation 70. "Used" is the solves a run used, where a censored entry (–) stops. Budgets: 190 evaluations; 114 for the first $\sigma_0 = 0.5$ seed on geode-fem, 1000 for the long run, 2,500 solves for finite differences, checked after each gradient. "Best" is over the counted evaluations, which exclude the start (−1.47 dB). Seeds: 18858188808230, 2 and 3 at $\sigma_0 = 0.5$; 11, 12 and 13 for the other CMA-ES runs.

Optimizer, setting	Forward	Used	Solves to			Best
			−9 dB	−9.5 dB	−10 dB	(dB)
L-BFGS, adjoint	geode-fem	125	30	35	60	−10.04
L-BFGS, finite differences	geode-fem	2,535	1,170	1,365	2,340	−10.04
sep-CMA-ES, $\sigma_0 = 0.5$	geode-fem	535/785/690	405/385/395	520/385/395	–/695/–	−9.56/−10.02/−9.77
same seeds	Palace	860/780/660	405/385/395	805/385/570	–/–/–	−9.58/−9.89/−9.90
sep-CMA-ES, $\sigma_0 = 1.0$	geode-fem	460	380	–	–	−9.29
sep-CMA-ES, $\sigma_0 = 1.5$	geode-fem	95/100/105	–	–	–	−6.29/−1.48/−1.30
1000 evaluations	geode-fem	1,865	–	–	–	−8.81
CMA-ES, $\sigma_0 = 0.5$	geode-fem	695	230	690	–	−9.82
CMA-ES, $\sigma_0 = 1.5$	geode-fem	80	–	–	–	−3.51
NEWUOA, $n + 2$ points	geode-fem	945	380	400	650	−10.06
NEWUOA, $2n + 1$ points	Palace	945	415	420	–	−9.95

The Palace CMA-ES runs reuse the three geode-fem seeds at $\sigma_0 = 0.5$; with the same seed the two solvers make the same evaluations to about evaluation 70, so those six runs are three trajectories, each under two forward models, which isolates the forward solver; seed 2 also reaches −9.5 dB at the same count under both.

Settings. Each optimizer was run at two or three values of its main setting (Table 2): σ_0 for CMA-ES and the number of interpolation points for NEWUOA; the best CMA-ES setting is the smallest σ_0 tried.

Counting solves. As defined in Section 1, one EM solve is one band frequency with one matrix factorization, and a level is a target for the worst-of-band $|S_{11}|$, here the worst of the five band points in a run's own forward model. The adjoint solve reuses the forward factorization, and the adjoint's start, which yields its first gradient, is counted. The adjoint run's recorded 190 driven solves (every forward solve the solver logged, the count Table 5 also uses) include 65 that are not counted, leaving the 125 of Table 2: 35 for the three gradient checks and a fresh evaluation of the reported design, and 30 for the verification sweeps (9 in-band and 21 wide points). The finite-difference control is charged 5 solves per guard-passing trial plus 190 per gradient. The gradient-free runs' separate evaluation at $\mathbf{c} = 0$ is not counted, except as NEWUOA's own first evaluation, and a sample rejected by the guard costs no solve. A level a run did not reach is censored at the solves it used.

Results with 38 parameters. To reach −9.5 dB the adjoint uses 35 solves (Table 2, Figure 3a). The fastest gradient-free run, a separable CMA-ES seed at $\sigma_0 = 0.5$, uses 385 (11×), NEWUOA with $n + 2$ points 400 (11.4×), and no run fewer. These ratios are against the fastest runs that reached the level; the $\sigma_0 = 1.5$ runs, censored at 80 to 105 solves, bound the −9.5 dB ratio only from below, by 2.3 to 3.0×. The separable CMA-ES is not cross-checked against a reference implementation, but NEWUOA, from PRIMA, gives 11.4× on its own. To −9 dB the ratio to the fastest run to reach it is 7.7×: full CMA-ES at $\sigma_0 = 0.5$ needs 230 solves against 30, in a single seed, and then

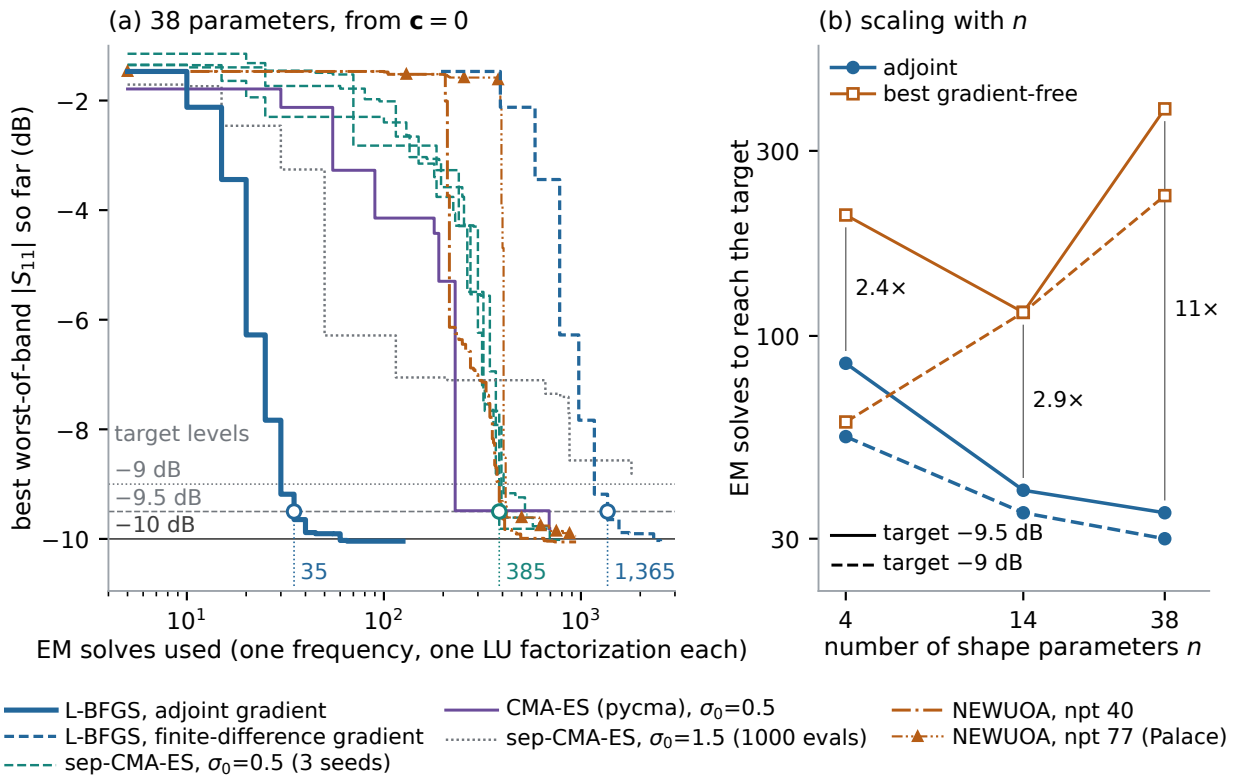


Figure 3: Reading solves to target off the optimization histories. (a) Best worst-of-band $|S_{11}|$ so far on the L1 optimization model against EM solves used, with 38 parameters, for the adjoint and finite-difference L-BFGS and the gradient-free runs of Table 2 (the Palace CMA-ES runs, which share seeds with the dashed geode-fem runs, the $\sigma_0 = 1.0$ run and the short $\sigma_0 = 1.5$ runs are omitted). The horizontal lines are the target levels of Table 2; a run's solves to target is where its curve first reaches a line. The open circles mark this for −9.5 dB: the adjoint at 35, the fastest gradient-free run (separable CMA-ES, seed 2) at 385 and the finite-difference control at 1,365. (b) Solves to −9.5 dB (solid) and −9 dB (dashed) against the number of parameters, for the adjoint and for the best of the 5 to 11 gradient-free runs at each n; the vertical bars and their labels give the ratio at −9.5 dB. Data: `tuned/`, `designc/c4/`, `v6/x4/`, `v6/palace_loop/`.

690 to reach −9.5 dB. A larger step does not help: at $\sigma_0 = 1.5$, 89 to 92% of the samples fail the volume guard, a 190-evaluation run makes only 16 to 21 feasible evaluations, and the seed given 1000 evaluations (1,865 solves) ends at −8.81 dB, so the guard bounds the useful step. The smaller NEWUOA interpolation set shortens the start-up but not the time to −9.5 dB (400 against 420 solves); that run also reaches −10.06 dB, the best level of any 38-parameter run. The $\sigma_0 = 0.5$ CMA-ES and NEWUOA runs had 4 to 8 times the adjoint's solves, so the budgets favour them.

At −10 dB the comparison is weak. Two gradient-free runs reach it, at 650 and 695 solves against the adjoint's 60, and the rest are censored. The forward models agree to 0.004 dB on the 41 designs both evaluated (the first CMA-ES generation of each seed, 42 samples with one rejected by the guard in both) but differ by up to about 0.25 dB at optimized designs (Palace reads the adjoint's final design at −9.80 dB against geode-fem's −10.04 dB), the size of the gap between reaching and missing −10 dB, so we derive no ratio from the −10 dB column.

The finite-difference control. With a forward-difference gradient, L-BFGS takes the adjoint's path. The two gradients agree to 4.6×10^{-7} (relative) at the start and 5.2×10^{-4} at the final design,

the worst-of-band value at each of the 12 steps matches the adjoint run's to 1.7×10^{-5} dB, and both reach -10.04 dB at step 12. A step costs 195 solves against the adjoint's 5, so the ratio is $39\times$ at every level, and the 1,365 solves to -9.5 dB are 2.6 to 3.55 times the 385–520 of separable CMA-ES at $\sigma_0 = 0.5$ and more than every gradient-free run that reached the level. With the optimizer, basis, guard and iterates fixed, only the cost of the gradient changes, so this ablation credits the solve savings to the adjoint; here a plain forward-difference gradient on 38 parameters is worth less than none. Broyden-type updates reduce finite-difference sensitivity cost in antenna optimization (Koziel and Pietrenko-Dabrowska, 2020), and trust-region microwave design has been run with adjoint and with finite-difference sensitivities (Koziel et al., 2012); we did not try a reduced-cost finite-difference scheme. The design itself comes from the optimizer and the parametrization, which give the same iterates with either gradient.

Scaling with the number of parameters. From the same cold start with 4 and 14 parameters, against the best of three separable CMA-ES seeds at $\sigma_0 = 1.5$ and NEWUOA with $n + 2$ and $2n + 1$ points, the ratio to the fastest run to reach each level is $1.1\times$ at -9 dB and $2.4\times$ at -9.5 dB with 4 parameters, $3.3\times$ and $2.9\times$ with 14, and $7.7\times$ and $11\times$ with 38 (Figure 3b). With 4 the adjoint stops on the guard at -9.85 dB and three gradient-free runs end lower (-9.92 to -9.97 dB); with 14 it reaches -10.00 dB in 60 solves and the best gradient-free run ends at -9.99 dB. The growth from 4 to 14 parameters comes from the adjoint getting faster with the richer basis (85, 40 and 35 solves to -9.5 dB at 4, 14 and 38), while the best gradient-free count is not monotone (205, 115, 385); only the step to 38 comes from gradient-free search slowing. Three problem sizes fix the direction of the trend but not its rate: from 14 to 38 parameters the best gradient-free count to -9.5 dB grows 3.3-fold for 2.7 times the parameters while the adjoint's falls slightly, but the 4-to-14 step moves the other way for gradient-free search, and we fit no law to three points from one problem. With 4 parameters gradient-free search is faster to -5 and -8 dB (5 and 45 against 30 and 55 solves), and the crossover is near -9 dB. At 4 and 14 parameters separable CMA-ES ran at $\sigma_0 = 1.5$ only, the worst setting at 38, which may favour the adjoint on the CMA-ES side, but NEWUOA, which has no σ_0, gives the best gradient-free count at both.

Why some gradient-free runs end lower. The adjoint's final level is set by step rejection at the volume guard: L-BFGS with a reject-the-step guard stops at the edge of the feasible set, while CMA-ES and NEWUOA rank or penalize infeasible samples and keep moving along it, which is why three gradient-free runs end lower with 4 parameters and why the -10 dB column is weak. A projected or penalized gradient step was not tried.

The forward solver and the wall-clock. The paired Palace runs reach -9 dB at the same counts as their geode-fem twins, so the result does not depend on geode-fem's forward model, though Palace's band evaluation is the cheaper one (64.4 against 72.1 s on an r6i.4xlarge). An adjoint band evaluation takes 95.1 s against 72.1 s for a forward one on the same host ($1.32\times$), so in geode-fem wall time the $11\times$ at -9.5 dB above is about $8.3\times$ (5,552 against 666 s). Table 3 collects the wall-clock costs of this paper's runs, each with the solves it spent and its host. To -9.5 dB the Palace-driven runs took a measured 84–172 min (CMA-ES) and 90 min (NEWUOA) against an estimated 11.1 min for the adjoint (its solves times the measured per-solve cost). These are serial times on one host; CMA-ES would spread across hosts, while L-BFGS spreads only over the five band frequencies, so solves to target is the hardware-independent measure.

Table 3: Wall-clock costs of the runs in this paper, each with the solves it spent and the host it ran on. "Counted" solves are those of Table 2; "recorded" solves are driven solves as logged (Table 5). An estimated time is counted solves times the per-solve cost measured on the r6i.4xlarge; the other times are measured.

Run	Solves	Host	Wall time
Adjoint L-BFGS to −9.5 dB, 38 parameters	35 counted	r6i.4xlarge rate	666 s (11.1 min), estimated
Fastest gradient-free run to −9.5 dB (separable CMA-ES, seed 2)	385 counted	r6i.4xlarge rate	5,552 s, estimated
Palace-driven separable CMA-ES to −9.5 dB, three seeds	385–805 counted	r6i.4xlarge, 16 MPI ranks	84–172 min
Palace-driven NEWUOA to −9.5 dB	420 counted	r6i.4xlarge, 16 MPI ranks	90 min
Finite-difference control, whole run	2,535 counted	16-vCPU host, 8 finite-difference jobs in parallel	4.7 h (1,164 to 1,211 s per gradient)
Final descent to design C (Table 5)	125 recorded	32-vCPU host	26 min
Palace $p = 3$ sweep of design C on L1	adaptive sweep	16 MPI ranks	3.5 h
Palace $p = 2$ sweep of design C on the 0.2 mm-rim mesh	adaptive sweep	16 MPI ranks	1.5 h
geode-fem direct solve on the combined mesh, one frequency	1	204 GiB peak memory	about 8 min

Table 4: Where the error sits: the initial patch's $|S_{11}|$ dip under refinement of L1, geode-fem at $p = 1$, with P2/L1 for reference. Each row is one refinement level. "Slab" is the element size in the substrate under the patch (Figure 1b), "rim" the mean edge length within 0.5 mm of the patch edge (Figure 1d), "layers" the median number of elements through the slab. Model-dependent values.

Mesh	Slab (mm)	Rim (mm)	Layers	Unknowns	Dip (GHz)	vs L1 (MHz)
L1	0.80	0.78	2.00	150,561	2.3118	0
slab 0.533	0.533	0.52	2.54	234,691	2.3273	+15.5
slab 0.40	0.40	0.39	3.44	414,837	2.3404	+28.6
slab 0.30	0.30	0.29	4.19	828,775	2.3480	+36.2
rim 0.40	0.80	0.39	1.99	317,424	2.3420	+30.2
rim 0.20	0.80	0.20	1.99	430,808	2.3569	+45.1
slab 0.30 + rim 0.10	0.30	0.10	4.19	1,738,557	2.3659	+54.1
L1 at $p = 2$ (P2/L1)	0.80	0.78	2.00	804,460	2.3566	+44.8

4 Forward-Model Accuracy

The error sits at the rim. Table 4 refines L1 in two ways. Refining the whole slab under the patch to 0.40 mm gives 3.4 layers and, as a side effect, a 0.39 mm rim. Refining only within 0.5 mm of the patch edge to the same rim size, with the 2-layer interior unchanged, lands within 1.6 MHz of it (2.3404 against 2.3420 GHz). The interior layer count does not decide the dip; the rim does, as expected where the field is singular at a conductor edge (Meixner, 1972). A 0.2 mm rim at $p = 1$ (431k unknowns) matches P2/L1 (805k) to 0.3 MHz. Over all seven $p = 1$ refinement levels of Table 4 the dip follows $f = f_\infty - C\, h_{\mathrm{rim}}^{0.80}$ with a 1.3 MHz rms residual (Figure 4b). The exponent is a fit to the rim edge length on graded meshes, not a proven asymptotic order. The combined level, which halves the rim to 0.1 mm and refines the slab, moves the dip a further 9.0 MHz beyond the 0.2 mm-rim level.

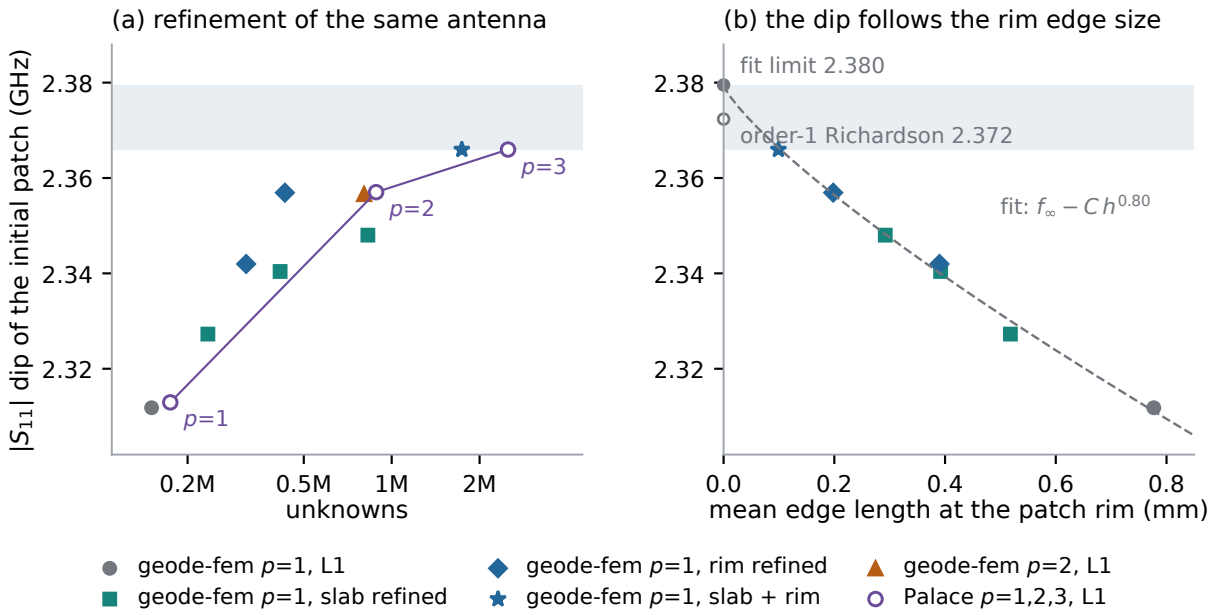


Figure 4: The initial patch's $|S_{11}|$ dip under refinement. (a) Against the number of unknowns: geode-fem at $p = 1$ on L1 and its slab, rim and combined refinements, geode-fem at $p = 2$ on L1, and Palace at $p = 1, 2, 3$ on the same L1 mesh. (b) Against the mean edge length at the patch rim, with the fit $f_\infty - Ch^{0.80}$ (dashed), its limit and the order-1 Richardson step on the 0.39/0.20 mm rim pair. The shaded band, 2.366–2.380 GHz, runs from the finest computed value to the fit's limit; it is an estimate, not a bound. Data: `v6/v6_summary.toml [x1]`.

An independent solver on the same meshes. Palace (Palace developers, AWS Center for Quantum Computing, 2023) solved the same problem on geode-fem's own meshes, written out with their physical tags, with the same materials, PEC surfaces and uniform 50 Ω lumped port. Its absorber differs (air in the UPML shell, a first-order absorbing outer wall); its driven sweeps use an adaptive reduced-order model with tolerance 10^{-3} on a 1 MHz grid, and GMRES to 10^{-8}. On L1, Palace and geode-fem agree on the dip to 1.1 MHz at $p = 1$ (2.3130 against 2.3118 GHz) and 0.4 MHz at $p = 2$ (2.3570 against 2.3566 GHz); on design A (Section 5) the P2/L1 band edges agree to 0.7 MHz (lower) and 0.2 MHz (upper). Palace at $p = 3$ on L1 gives 2.3660 GHz, 8.9 MHz above its $p = 2$ value, and geode-fem's finest refinement level (slab 0.30 plus rim 0.10 mm) gives 2.3659 GHz (Figure 4a): two different refinements of the same model, order in Palace and rim size in geode-fem, land within the 1 MHz sweep step. Sharing mesh, port model and geometry, this checks the discretization, not the model.

An FDTD cross-check. A Meep (Oskooi et al., 2010) model of the initial antenna checks the resonance with an independent method. It puts zero-thickness PEC sheets on the Yee grid, keeps the 1.6 mm gap in every column (4 cells at 0.4 mm, 8 at 0.2 mm), uses a port of exactly 50.0 Ω and models the FR-4 loss as a conductivity ($\tan\delta = 0.0195$–0.0205 over 2.30–2.42 GHz); an earlier Meep model with one-cell metal shells, which give the normal and tangential field different gaps on the curved board, placed the bent dip about 250 MHz low.

The flat board dips at 2.3683 GHz, 10.3 MHz above geode-fem's finest flat model ($p = 2$ on L1, itself about 9 MHz below $p = 3$ for the bent patch), and the bend moves the dip by −1.6 MHz, against geode-fem's −1.0 to −1.6 MHz. The bent dip is at 2.3667 GHz with 0.4 mm cells and

2.3611 GHz with 0.2 mm cells (Re Z 44.8 and 47.3 Ω), against 2.3660 GHz for Palace $p = 3$ and for geode-fem's finest refinement level. Meep's dips are parabola vertices through samples 10 MHz apart, so differences of about 1 MHz are below their resolution. The two Meep grids also differ in patch length (28.8 against 29.0 mm), width (37.2 against 37.6 mm, nominal 38) and probe inset (7.0 against 6.9 mm), so their −5.6 MHz change is a trend, not a mesh-limit value. The refined models of the three codes place the bent dip between 2.3611 and 2.3667 GHz.

The mesh-limit resonance. Every refinement computed so far raises the dip. If that continues, the finest computed value, 2.366 GHz, is a lower estimate of the mesh-limit dip; this is an inference from the trend, not a proof. The rim fit puts the limit at 2.380 GHz and the order-1 Richardson step on the rim pair at 2.372 GHz; since the fitted exponent is below 1, the order-1 step is the lower of the two if the fit holds. We therefore estimate the initial patch's mesh-limit dip at 2.366–2.380 GHz: the floor is the finest computed value, and the two extrapolations put the limit 6.4 MHz (Richardson) to 13.5 MHz (rim fit) above it, the range that Table 7 carries.

Port-free resonance. Palace's eigenmode solver, on the same L1 mesh at $p = 2$ with the port gap left open (no feed), finds the patch mode at 2.3408 GHz with $Q = 22.2$ for the initial patch, and at 2.3991 GHz with $Q = 19.3$ for design C of Section 5 (7 modes, backward error below 5×10^{-9}), identified by its substrate energy fraction (0.80–0.83, every other mode at most 0.013). The matched dip on the same discrete model lies above it, by 16.3 MHz for the initial patch and 30.9 MHz for design C. In this model the port is a resistive sheet: loading the gap with it raises the mode by 12.2 and 23.1 MHz and halves its Q (11.5 and 8.6), and the driven dip sits a further 4.1 and 7.7 MHz above the loaded mode. A series inductance, as in a coaxial probe, would move the match the same way. The Hammerstad transmission-line estimate for the initial patch, 2.433 GHz (Hammerstad, 1975), lies 92.5 MHz above the port-free resonance at $p = 2$ and 76 MHz above the matched dip. The discretization accounts for 9 to 22 MHz of that and the feed moves the dip toward the closed form, so most of the gap is in the cavity model; the finite ground and the closed form's own accuracy are untested candidates.

Absorber. Over a ladder of UPML conductivity and thickness, $|S_{11}|$ changes by at most 0.021.

5 Design Procedure

Frequency space mapping. A design optimized on L1 resonates higher on a finer model. We model the fine response as the coarse response shifted in frequency, $S_{\text{fine}}(f) \approx S_{\text{coarse}}(f - \delta)$, fit δ by least squares over the fine model's points, and re-optimize on the coarse model for the band shifted down by δ. For the cold L1 optimum of Section 3 the fitted L1-to-P2/L1 shift is 47.85 MHz (complex misfit 0.23–0.38 unshifted, 0.018–0.025 shifted); without it, the cold L1 optimum (−10.04 dB on L1) gives −4.76 dB at P2/L1. The design is built by a chain of such descents, each warm-started from the previous design with the shift and, where needed, the mesh updated; we call this the space-mapping chain. Its last descents give the three designs of Table 5. The shift is re-fitted at each descent as the design moves: the cold optimum's 47.85 MHz had become 44.94 MHz by the last descents (Table 5).

Feed re-basing. Moving the feed toward the $+y$ edge compresses the mesh between them, and the cold runs stop there at the volume guard. We then regenerate the mesh with the port at its moved position, so that the moved feed becomes the unmorphed base of a new mesh (re-basing),

Table 5: The designs. δ is the band shift of the final descent. "Solves" counts the driven solves of that descent as recorded, including its gradient checks, a different count from Table 2's. "Fine model" is the model the design was centred for.

Design	Feed	δ (MHz)	Steps	Solves	Fine model
A	4.27 mm strip (free width)	44.94	9	125	P2/L1
B	1.6 mm probe (pinned)	44.94	10	130	P2/L1
C (final)	1.6 mm probe (pinned)	57.0	10	125	refined rim, Palace $p = 3$

and carry the design over by shifting the two feed coefficients. After re-basing, the optimizer widens the patch until the guard binds in the air 5.5 mm outside a $\pm s$ edge, which is a non-radiating edge.

Three designs. Table 5 lists the designs. Design A ends the space-mapping chain centred for P2/L1 with the feed width free, which widened the port into a 4.27 mm strip. At P2/L1 it reaches −10.78 dB with a 94.9 MHz band, 2.3938–2.4887 GHz. At Palace $p = 3$ its band moves up to 2.4029–2.4975 GHz, the lower edge misses 2.400 GHz by 2.9 MHz, and the worst in-band value is −9.56 dB, so a design centred on P2/L1 misses once the rim is resolved. Design B re-runs A's last step with the probe width pinned at 1.6 mm and reaches −10.62 dB at P2/L1 (−10.65 dB with a 1.27 mm probe), so the wide strip was not what carried the bandwidth.

Design C re-centres B for the refined model with a band shift of 57.0 MHz. The shift adds three terms: 44.94 MHz from L1 to P2/L1, 8.7 MHz for design A's $p = 2$ to $p = 3$ dip shift in Palace, and 3.4 MHz further down because every refinement so far has moved the band up.[1] Because Palace $p = 3$ informed both the shift and the choice between the two runs, the geode-fem mesh that was not used in it carries the independent check in Section 6. The descent to C ran 10 steps and 125 solves, from −8.73 to −10.64 dB on its shifted L1 band, and stopped on the guard with the gradient norm down from 0.143 to 2.2×10^{-3}; the gradient matched finite differences to 1.2×10^{-6}. As built, C is 29.00 mm long and 49.72 mm wide on the centre line, with a barrel outline whose $\pm s$ edges bow out most at mid-length, and its 1.6 mm probe sits 5.14 mm from the $+y$ edge.

6 Results

On the finest models. On Palace at $p = 3$ on L1, design C has a worst in-band $|S_{11}|$, over the nine points of Table 6, of −10.46 dB, at 2.4835 GHz, and a contiguous −10 dB band from 2.3935 to 2.4864 GHz, 93.0 MHz wide, which covers the 83.5 MHz ISM band with 6.5 MHz to spare at 2.400 GHz and 2.9 MHz at 2.4835 GHz (Table 6, Figure 5a). geode-fem's finest mesh, the combined 0.1 mm rim and 0.3 mm slab (1.74M unknowns), gives −10.73 dB and a 94.7 MHz band, 2.3934–2.4881 GHz. The two lower edges agree to within the sweep resolution (the interpolated values differ by 0.02 MHz); the upper edges differ by 1.7 MHz, partly because geode-fem's upper crossing is interpolated across a 5 MHz gap. Palace at $p = 2$ on the 0.2 mm-rim mesh (2.47M unknowns), the most refined model computed for the design, gives −10.32 dB and a band of 2.3981–2.4912 GHz: its lower edge sits 4.6 MHz above Palace $p = 3$ on L1, which leaves 1.9 MHz of lower-edge margin. The two models are different refinements of comparable size (one more order on L1, and the rim at $p = 2$), so the 4.6 MHz shows that $p = 3$ on L1 does not yet resolve the rim. From $p = 2$ to $p = 3$ on L1 the design's dip rises 9.1 MHz and its lower band edge 8.7 MHz; the initial patch's dip rises 8.9 MHz over the same step, so the design shifts like the patch.

[1] A parallel run with the first two terms only, 53.6 MHz, was set aside by a prediction from the fine models. For each run we added the initial patch's step from the 0.2 mm-rim mesh to Palace $p = 3$, about 9 MHz, to its lower edge on that mesh; this predicts a lower-edge margin at $p = 3$ of about 2.6 MHz for the 53.6 MHz run and about 6 MHz for C.

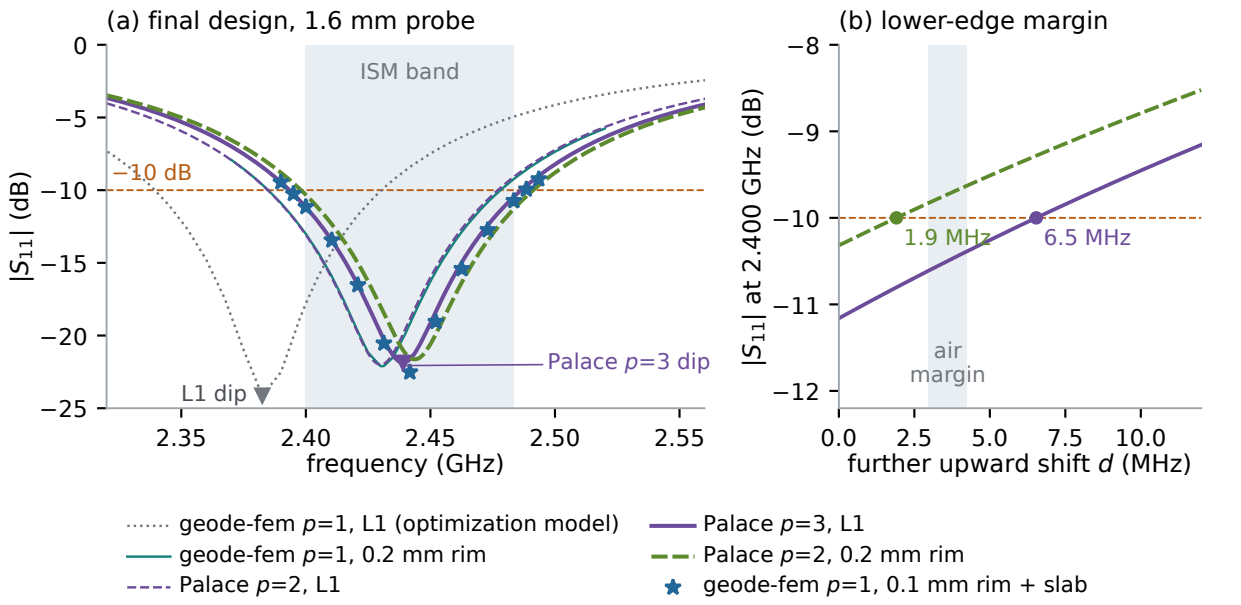


Figure 5: (a) Return loss of design C. Palace at $p = 3$ and $p = 2$ on the design's L1 mesh and at $p = 2$ on its 0.2 mm-rim mesh (1 MHz grids); geode-fem at $p = 1$ on the combined 0.1 mm-rim and 0.3 mm-slab mesh (13 points) and on the 0.2 mm-rim mesh; and geode-fem at $p = 1$ on L1, the optimization model, where the design sits 57 MHz low by construction. Triangles mark the dips of Palace $p = 3$ and of L1; the dashed line is the −10 dB target and the shading the ISM band, as in Figure 2. (b) $|S_{11}|$ at 2.400 GHz if a whole Palace response moves up by a further d, read from the computed response at $2.400 - d$ GHz; the dots mark the lower-edge margins, 6.5 and 1.9 MHz. The shaded range is the rise of the lower edge with a 60 or 80 mm air margin (geode-fem, $p = 1$, L1); the budget of Table 7 uses the dip's 4.3 ± 1.3 MHz. Data: `recentre/`, `designc/c3/`, `tuned/tuned_summary.toml`.

Table 6: Design C on each model, in order of refinement, with design A on Palace $p = 3$ for comparison. "Worst" is over the nine in-band points (2.400 to 2.4835 GHz in 10.4375 MHz steps). Band crossings are interpolated linearly in dB. Margins are 2.400 GHz minus the lower crossing and the upper crossing minus 2.4835 GHz. Design C's meshes are the re-based fixtures with the port at its moved position, so their counts differ slightly from Table 4's; Palace counts 80,200 more unknowns at $p = 2$ on the same mesh because it duplicates the vertices on interior PEC and port surfaces. Model-dependent values.

Model	Unknowns	Worst (dB)	−10 dB band (GHz)	Coverage	Margins (MHz)
geode-fem $p = 1$, L1	150,541	−4.94	2.3393–2.4304	36%	60.7 / −53.1
geode-fem $p = 2$, L1	804,336	−9.08	not swept	not swept	not swept
Palace $p = 2$, L1	884,536	−9.08	2.3847–2.4770	92.2%	15.3 / −6.5
geode-fem $p = 1$, rim 0.2	431,528	−9.23	2.3849–2.4782	93.6%	15.1 / −5.3
geode-fem $p = 1$, rim 0.1 + slab	1,738,279	−10.73	2.3934–2.4881	100%	6.6 / 4.6
Palace $p = 3$, L1	2,503,077	−10.46	2.3935–2.4864	100%	6.5 / 2.9
Palace $p = 2$, rim 0.2	2,465,984	−10.32	2.3981–2.4912	100%	1.9 / 7.7
Design A, Palace $p = 3$	2,729,331	−9.56	2.4029–2.4975	96.5%	−2.9 / 14.0

Air margin and curvature. Re-meshing design C with a 60 or 80 mm air margin instead of 40 mm, at $p = 1$ on L1, moves its dip by 4.3 MHz at both (4.34 and 4.31 MHz), so the resonance has levelled off by 60 mm. The band edges scatter about that between the three air meshes: the lower edge rises 3.0 and 4.3 MHz, the upper 4.2 and 2.9 MHz, and the −10 dB bandwidth is 91.0, 92.3 and

Table 7: What moves design C's lower −10 dB edge, against its margin on Palace $p = 3$, L1. A negative entry uses up margin.

Term	MHz	Tier, source
Lower-edge margin, Palace $p = 3$, L1	+6.5	model-dependent, Table 6
Rim refinement at $p = 2$ (Palace, 0.2 mm rim)	−4.6	model-dependent, Table 6
Air margin, 40 → 60 / 80 mm (dip; edges scatter by 1.3)	−4.3	model-dependent at $p = 1$ on L1, level by 60 mm; transfer estimated
Further refinement above the 0.2 mm-rim model	not computed	initial patch: 6.4–13.5 above Palace $p = 3$, estimated
Coaxial feed in place of the lumped port	not quantified	
Copper loss in place of PEC	not quantified	

89.7 MHz. The air-margin term is therefore about 4.3 MHz with about 1.3 MHz of re-meshing scatter in the edges, smaller than the refinement shifts but of the size of the design's margins. Design A's copper (the strip-fed design, Table 5) on a flat board (bend radius 10^6 mm, the same mesh node for node) resonates 0.3 MHz from the bent one at $p = 1$, and a complex frequency-shift fit at P2/L1 finds 0.28 MHz. The bend does not detune that design, but it changes its match: the bent board has a 93.3 MHz $p = 1$ −10 dB band against 82.8 MHz flat, and a worst in-band $|S_{11}|$ at P2/L1 of −10.78 against −9.93 dB. Design C was not run flat, and we did not investigate the mechanism.

The lower-edge budget. Table 7 collects what moves the lower edge. We take the air-margin term as the dip's 4.3 MHz and carry the 1.3 MHz edge scatter between re-meshed air boxes (above) as the uncertainty of the measured terms; it is the scatter of the air re-meshes only, applied once to the sum. Added, they take more than the margin: $6.5 - 4.6 - 4.3 = -2.4 \pm 1.3$ MHz, a lower edge near 2.402 GHz, and on the 0.2 mm-rim response moved up by 4.3 MHz, 2.400 GHz sits at −9.63 dB (Figure 5b). The sum assumes that the terms add, and the air-margin term was measured on the coarse model. The rim step (4.6 MHz) is below both drift estimates of the initial patch above Palace $p = 3$ (order-1 Richardson 6.4 MHz, rim fit 13.5 MHz), so it need not be the last of them; Palace $p = 3$ on the 0.2 mm-rim mesh was not run. If the estimates transfer and include the rim step, up to $13.5 - 4.6 = 8.9$ MHz more remains. The estimated miss is therefore $2.4 - 1.3 = 1.1$ MHz to $2.4 + 1.3 + 8.9 = 12.6$ MHz, which we state as 1 to 13 MHz. A further re-centring cannot cover the whole range: the 93.0 MHz band is 9.5 MHz wider than the ISM band, less than the 11.5 MHz spread of the estimate, so the bandwidth, which the guard and the board limit, binds.

What is claimed. Measured on stated discrete models (model-dependent): design C's −10 dB band covers 2.400–2.4835 GHz on the three most refined models computed, Palace $p = 3$ on L1, geode-fem's combined 0.1 mm-rim and 0.3 mm-slab mesh and Palace $p = 2$ on the 0.2 mm-rim mesh, with 6.5, 6.6 and 1.9 MHz to spare at 2.400 GHz. Estimated: a larger air margin and further refinement each move the band up, and the budget above puts the mesh-limit lower edge 1 to 13 MHz above 2.400 GHz. The claim is therefore ISM coverage on those three discrete models. We do not claim ISM coverage for the mesh-limit model, whose lower edge we estimate misses the band, or for a built antenna, whose coaxial feed and copper loss are not modelled (Section 8).

Where the descent stopped. The descents to A, B and C stop on the volume guard. For C the gradient norm fell 65-fold first, so C may be near a stationary point of the guarded problem, and the widened patch comes within about 6 mm of the 62 mm board edge. Remeshing at the widened

outline would relieve the guard; it was not run. Given the 38-parameter chain's final band shift and feed re-base in advance (Section 5), a design in the four-parameter basis of Section 2.4 does reach −10.33 dB at P2/L1, against −10.78 dB for the 38-parameter design; its gradient norm grew (0.023 to 0.074) where C's fell.

Radiation efficiency and gain. For design C we computed the accepted port power, the Poynting flux through a closed surface in the air box, the FR-4 loss and the broadside gain by a near-to-far-field transform at $p = 1$, on L1 at the band it was optimized on and on the 0.2 mm-rim mesh across the ISM band. The power balance closes to 0.8% on L1 and 1.6% on the rim mesh. The radiation efficiency is 0.61–0.63, the dielectric loss fraction 0.37, and the broadside gain 4.45–4.66 dBi with a front-to-back ratio of 13–15 dB. With each model's own match the realized broadside gain is 4.06–4.50 dBi on L1 over its optimization band and 4.11–4.56 dBi on the rim mesh over the ISM band (model-dependent). With perfectly conducting (PEC) conductors this efficiency counts radiation against dielectric loss only; copper loss would lower it and the gain.

Bandwidth against a closed form. On the voltage-standing-wave-ratio-2 criterion, Palace gives design C a bandwidth of 99.0 MHz at $p = 3$ (design A: 100.3 MHz at $p = 2$ and 101.0 MHz at $p = 3$). The closed-form formulas of Jackson and Alexopoulos (1991), evaluated for design A's 49.75 × 28.80 mm rectangle (C's outline is 49.72 × 29.00 mm), give 70.4 MHz, and 67.9 MHz at the area-equivalent width of 45.4 mm. Since $\tan\delta$ is the only loss, design C's dielectric loss fraction (0.367–0.372) gives the model inequality $Q_t \geq 0.3667/\tan\delta = 18.3$ (not a bound in the sense of Section 1), against the closed form's $Q_t = 24.5$ and radiation Q of 48.0 for design A's rectangle. We read this, as an interpretation, as the model patch radiating more strongly than the infinite-ground closed form, perhaps through its finite 62 × 53 mm ground, which we did not test.

7 Related Work

Finite-element S-parameter sensitivities with ports. Design sensitivities of the scattering matrix for tetrahedral edge elements were derived by Webb and Akel (2000), extended to the frequency response by Webb (2002) and used in gradient optimization of microwave devices by Nair and Webb (2003). Georgieva et al. (2002) give adjoint S-parameter sensitivities that need only the solver's field solutions, so that commercial solvers can supply them; adjoint and self-adjoint formulations are reviewed by Nikolova et al. (2004a) and Nikolova et al. (2006b), and continuum shape sensitivity for high-frequency devices is given by Choi et al. (2011). HFSS computes S-parameter derivatives with respect to geometric parameters by reusing the LU factors (Ansys, Inc., 2025; Vardapetyan et al., 2008), and COMSOL documents adjoint shape optimization of waveguide devices on a deformed mesh (COMSOL AB, 2023; Jensen, 2022). For antennas, Ghassemi et al. (2013) evolve a microstrip antenna with an FEM adjoint sensitivity and a small set of control vertices. Toivanen et al. (2009) obtain antenna shape sensitivities from a method-of-moments solver by automatic differentiation, and Toivanen et al. (2010) optimize ultra-wideband antennas whose outlines are parametrized by splines, a close prior combination to our spline outline and gradient descent. Jacobsson and Rylander (2010) apply gradient-based shape optimization to conformal array antennas.

Gradient-free and surrogate-assisted antenna optimization. Evolutionary search is the common gradient-free route in antenna design, and surrogate models cut its cost: Liu et al. (2014) combine differential evolution with Gaussian-process surrogates, Zhou et al. (2021) use trust-region parallel Bayesian optimization, and Akinsolu et al. (2020) review machine-learning-assisted methods.

The closest precedent for our measurement is aerodynamic: Zingg et al. (2008) compare a genetic algorithm with an adjoint method on aerodynamic shape problems and find that the genetic algorithm needs 5 to 200 times as many function evaluations. We did not run a surrogate-assisted or Bayesian optimizer.

Variable-fidelity antenna optimization and space mapping. Space mapping aligns a cheap coarse model with an accurate fine model and optimizes the coarse one (Bandler et al., 1994, 2004; Koziel et al., 2006). In antenna design the coarse model is often the same electromagnetic solver on a coarse discretization, corrected by frequency scaling or by response-shape correction (Koziel, 2010; Koziel and Ogurtsov, 2014), and adjoint sensitivities have been used inside trust-region space mapping (Koziel et al., 2013). Our correction is the simplest member of this family: a frequency shift fitted between two discretizations of the same solver.

Automated shape differentiation and structured grids. Shape derivatives of finite-element models can be generated automatically in UFL and Firedrake (Dokken et al., 2020; Ham et al., 2019; Paganini and Wechsung, 2021) and in NGSolve (Gangl et al., 2021). Wang and Anderson (2011) give a discrete Maxwell shape adjoint on unstructured meshes in two dimensions, in the time domain and without ports. On structured grids, conductors are designed by density-based topology optimization (Erentok and Sigmund, 2011; Hassan et al., 2014a,b) and with FDTD shape sensitivities (Nikolova et al., 2004b, 2006a). A plain Yee grid staircases a curved conductor (Cangellaris and Wright, 1991; Häggblad and Runborg, 2014), which locally conformal FDTD avoids (Benkler et al., 2006; Dey and Mittra, 1997; Mittra and Yu, 2000; Zagorodnov et al., 2003). Subpixel-smoothed projection gives the Meep density adjoint subpixel-accurate boundaries (Hammond et al., 2025; Romano et al., 2026), though its material-grid adjoint excludes metals (Meep developers, 2026); Tidy3D offers autograd shape gradients and conformal PEC averaging (Flexcompute, Inc., 2026); shape parameters can be mapped to a fixed grid (Hooten et al., 2025; Liu and Bonar, 2024); and boundary-integral methods carry recent PEC shape-sensitivity work (Arens et al., 2026; Balouchev et al., 2024; Takahashi, 2025).

Conformal and cylindrical patch antennas. The cylindrical-rectangular patch has cavity-model analyses (Krowne, 1983; Luk et al., 1989), and nonplanar microstrip antennas and conformal arrays are treated in the monographs of Wong (1999) and Josefsson and Persson (2006). We use a full-wave model and do not compare with these analyses; the curvature here is mild (Section 6).

8 Limitations and Future Work

Feed, losses, refinement and scope. The probe is a uniform lumped port of the pin's width across the substrate, not a coaxial probe. A coaxial feed adds the pin's series inductance and a coax-to-substrate transition. Loading the gap with the lumped port alone moves design C's mode by 23 MHz, so a coax-fed model could move the match by megahertz, as much as the margins; Palace supports wave ports, and a coax-fed model was not run. Conductors are PEC, and the design has not been fabricated or measured. The drift above the 0.2 mm-rim $p = 2$ model is estimated, not computed. A $p = 2$ shape adjoint, or optimizing on a rim-refined mesh, would remove the coarse-to-fine shift that space mapping corrects here, and re-centring with the air-margin term included is the next step. The optimizer comparison is one problem from one start on the coarse model, with one full-CMA-ES seed per step size and one NEWUOA trust radius; a surrogate-assisted or Bayesian optimizer may narrow the gap, and whether the adjoint helps at the refined level is not tested.

9 Conclusion

On the copper design of a bent 2.4 GHz patch, a finite-element shape adjoint reaches a worst-of-band $|S_{11}|$ of -9.5 dB on the coarse optimization model in 35 EM solves, where the fastest gradient-free run needs 385 (11×, or 8.3× in geode-fem wall time), and -9 dB in 30 against 230 (7.7×). The ratio at -9.5 dB is 2.4×, 2.9× and 11× with 4, 14 and 38 parameters: it rises with their number, first because the adjoint gets faster with a richer basis and then because gradient-free search slows, and three sizes of one problem fix no rate. A finite-difference gradient on the same path costs more than gradient-free search, so the savings are the adjoint's. The optimum is no better, since the guard stops this L-BFGS where gradient-free search reaches the same level given enough solves. What the adjoint buys is solves, not a better optimum, and it buys them on the coarse model, so the design still has to be carried to finer models by a fitted frequency shift and verified there. Carried that way, re-centred using Palace at third order and a rim-refined mesh, design C covers the ISM band at -10 dB on the three most refined models computed, with 1.9 MHz to spare on the most refined; that coverage is what we claim (Section 6). The mesh-limit design is estimated to miss 2.400 GHz by 1 to 13 MHz; a $p = 2$ adjoint and a re-centring that includes the air-margin term (Section 8) are the next steps.

Acknowledgments. The authors' contributions and the AI-orchestrated development of geode-fem and of this paper are described in Section 10.

10 How geode-fem Was Designed and Built

geode-fem implements the whiteroom operator specification (Dunnavant, 2026), whose four layers restate Palace's cited source as pure functions, fused primitives, tensor operations and a small graph-evaluation calculus. C.D. wrote that specification, gave design input on the solver's operator layer, contributed code (the migration to Rust 2024, the Metal (Apple GPU) backend, the present layout of the core crate and the examples, and the geode-util crate) and reviewed this paper. The rest of the code, the experiments and the text of this paper were produced by AI models working as role-separated agents: Anthropic's Claude (Opus, Fable and Sonnet models, through Claude Code) and OpenAI's GPT Sol and GPT Astra models (via Codex). The agents were coordinated by Loom (Walters, 2025), an issue, worktree and pull-request workflow, and by Squad, a message room the agents share. Nearly every change was reviewed by a separate agent session. The agent reviews of the experiments behind this paper recomputed the reported numbers from the archived records, and the script that checks this paper's text against those records was mutation-tested. R.W. directed the research, ran the agents, made every scoping and claim decision, and takes responsibility for the results and the content.

11 Reproducibility

Every result in this paper is read from an archived, scripted artifact.[2] Table 8 lists them, relative to `benchmarks/patch_antenna_conformal/`. Each run record holds the exact command and the SHA-256 of its mesh; the meshes are regenerated from YAML recipes in `reference/gmsh/` with gmsh 4.15.2, and runs on re-based fixtures need the `--feed-support-legacy` flag. Many records carry no generating commit (they ran outside a git checkout); the code is merged at geode-fem commit `99646f06`, and the setting sweep and air-margin runs used `dcd8df60`. Palace is commit `fba6a5b`, built from `reference/palace/docker/Dockerfile`; PRIMA was built from libprima

[2]The geode-fem source is public at `https://github.com/rjwalters/geode-fem`. This paper's PDF and LaTeX source are archived at doi:10.5281/zenodo.21480892.

Table 8: Artifacts behind each result, relative to `benchmarks/patch_antenna_conformal/` except where a path begins with `benchmarks/`, `reference/` or `crates/` (repository root).

Result	Artifact
Fixture, refinement, Palace cross-check, curvature (Sections 2.1, 4, 6)	`baseline_2g4_L1.toml`, `convergence.toml`, `upml_ladder_L1.toml`; `v6/v6_summary.toml`, `v6/x1/`, `v6/x2/`
Baselines and scaling (Section 3)	`tuned/tuned_summary.toml`, `tuned/`; `v6/palace_loop/palace_loop_summary.toml`, `v6/x4/`; `copper/L1_copper_38.toml`; `designc/c4/`
Designs A to C (Sections 5, 6)	`copper/g2_summary.toml`, `v6/x3/`, `v6/x5/`; `recentre/recentre_summary.toml`, `recentre/palace/`; `designc/designc_summary.toml`, `designc/c1/`–`c3/`; `tuned/b5/`
FDTD cross-check (Section 4)	`reference/meep/docker/DIAGNOSIS.md`, `reference/meep/docker/HARNESS_V2.md`; `benchmarks/fdtd_density_baseline/meep_runs/v2/ladder.json`
Generator and tests	`crates/geode-core/examples/patch_conformal_nested_dof.rs`; `crates/geode-core/tests/patch_conformal_*.rs`

commit `43863c69` (package `0.7.3.dev1801+g43863c69c`), and pycma is 4.5.0. geode-fem ran on AWS Graviton4 hosts (r8g.4xlarge and r8g.8xlarge), Palace and the cost measurements on r6i.4xlarge. The figure scripts in `figures/src/` read the artifacts, `provenance.md` maps every number to its source field, and `verify_evidence.py` checks the text against them.

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